



JABchem



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Past Papers Nat 5 Chemistry

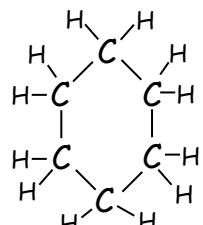
2014 Marking Scheme

Grade Awarded	Mark Required (/100)	% candidates achieving grade
A	69+	29.5%
B	58+	23.7%
C	48+	19.7%
D	43+	7.9%
No award	<43	19.3%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	13.9 /20	32.1 /60	12.3 /20

2014 National 5 Chemistry Marking Scheme

MC Qu	Answer	% Pupils Correct	Reasoning
1	A	91	$\text{rate} = \frac{\Delta \text{quantity}}{\Delta \text{time}} = \frac{60-0}{20-0} = 3 \text{ cm}^3 \text{ s}^{-1}$
2	D	83	☒ A Molecule contains 2xH, 1xBr and 1xCl ∴ molecule is not chiral ☒ B Molecule contains 3xH and 1xBr ∴ molecule is not chiral ☒ C Molecule contains 2xH, 1xI and 1xCl ∴ molecule is not chiral ☒ D Molecule contains 1xH, 1xBr, 1xI and 1xCl ∴ molecule is chiral
3	A	51	Phosphate PO_4^{3-} is listed in data booklet page 8 ∴ Total negative charge in $\text{Zn}_3(\text{PO}_4)_2$ formula is 6- as there are two phosphate ions All ionic compounds are neutral over all so the total positive charge must be 6+ ∴ Total positive charge in the 3 zinc ions = 6+ ∴ Positive charge on each zinc ion = $6+/3 = 2+$
4	C	89	$\text{Fe}_2\text{O}_3 + \text{CO} \rightarrow \text{Fe} + \text{CO}_2$ Fe: 2xFe before arrow but 1xFe after arrow ∴ double Fe after arrow $\text{Fe}_2\text{O}_3 + \text{CO} \rightarrow 2\text{Fe} + \text{CO}_2$ O: 4xO before arrow but 2xO after arrow ∴ double CO_2 after arrow $\text{Fe}_2\text{O}_3 + \text{CO} \rightarrow 2\text{Fe} + 2\text{CO}_2$ C: 1xC before arrow but 2xC after arrow ∴ double CO before arrow $\text{Fe}_2\text{O}_3 + 2\text{CO} \rightarrow 2\text{Fe} + 2\text{CO}_2$ O: 5xO before arrow but 4xO after arrow ∴ increase CO to 3 before arrow $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 2\text{CO}_2$ C: 3xC before arrow but 2xFe after arrow ∴ increase CO_2 to 3 after arrow $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$
5	C	75	☒ A All aqueous solutions contain both hydrogen and hydroxide solutions ☒ B All aqueous solutions contain both hydrogen and hydroxide solutions ☒ C Acids contain more hydrogen ions than hydroxide ions ☒ D Alkalis contain more hydroxide ions than hydrogen ions
6	A	59	☒ A calcium oxide is a soluble metal oxide which dissolves to form an alkali ☒ B nickel oxide is a non-soluble metal oxide so has no effect on pH ☒ C nitrogen dioxide is a soluble non-metal oxide and dissolves to form an acid ☒ D sulphur dioxide is a soluble non-metal oxide and dissolves to form an acid
7	D	82	☒ A formed by the neutralisation of calcium hydroxide and nitric acid ☒ B formed by the neutralisation of sodium hydroxide and hydrochloric acid ☒ C formed by the neutralisation of potassium hydroxide and sulphuric acid ☒ D magnesium hydroxide cannot be made by the neutralisation of an acid
8	A	88	Spectator ions appear chemically unchanged on both sides of a chemical equation: $\text{H}^+_{(\text{aq})} + \text{NO}_3^-_{(\text{aq})} + \text{K}^+_{(\text{aq})} + \text{OH}^-_{(\text{aq})} \rightarrow \text{K}^+_{(\text{aq})} + \text{NO}_3^-_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})}$ $\xrightarrow{\text{K}^+ \text{ appears on both sides of equation}}$ $\xleftarrow{\text{NO}_3^- \text{ appears on both sides of equation}}$

9	C	87	Cyclohexane is a 6 carbon cycloalkane with a ring of carbons inside it 																																																																
10	B	83	☒ A longest chain in structure is 4 carbons ∴ name must end in butane ☒ B 2-methylbutane has 4 carbons in main chain and methyl CH₃- group on C₂ ☒ C methyl CH₃- group is on C₂ from right hand side ∴ 2-methyl at start of name ☒ D 2-methylpentane would contain 6 carbons in total																																																																
11	C	77	☒ A this would be a comparison of side group position on octane number ☒ B there is no comparison to be made with this selection ☒ C butane, pentane and hexane would be a valid comparison of chain length ☒ D there is no comparison to be made with this selection																																																																
12	B	40	☒ A reaction would form 1-bromobutane and 2-bromobutane ☒ B reaction would form 2-bromobutane <i>only</i> ☒ C reaction would form 1-bromopentane and 2-bromopentane ☒ D reaction would form 2-bromobutane and 3-bromobutane																																																																
13	C	88	<table><tr><td>Alcohol</td><td>Propan-1-ol</td><td>Propan-2-ol</td><td>Butan-1-ol</td><td>Butan-2-ol</td></tr><tr><td>Boiling Point</td><td>97°C</td><td>82°C</td><td>118°C</td><td>100°C</td></tr></table>	Alcohol	Propan-1-ol	Propan-2-ol	Butan-1-ol	Butan-2-ol	Boiling Point	97°C	82°C	118°C	100°C																																																						
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14	C	47	☒ A energy being required to start a reaction does indicate either exo/endothermic ☒ B heat being given off indicates an exothermic reaction ☒ C A temperature drop during a reaction indicates an endothermic reaction ☒ D A temperature rise during a reaction indicates an exothermic reaction																																																																
15	A	75	<table><tr><th>Metal</th><th>Potassium</th><th>Sodium</th><th>Lithium</th><th>Calcium</th><th>Magnesium</th><th>Aluminium</th><th>Zinc</th><th>Iron</th><th>Tin</th><th>Lead</th><th>Copper</th><th>Mercury</th><th>Silver</th><th>Gold</th><th>Platinum</th></tr><tr><td>Reaction With Oxygen</td><td colspan="8">Burn In Oxygen to Form Metal Oxide</td><td colspan="2">Slowly React With Oxygen</td><td colspan="5">No Reaction With Oxygen</td></tr><tr><td>Reaction With Water</td><td colspan="4">Fast Reaction With Water</td><td colspan="4">Slow Reaction With Water</td><td colspan="2">Faster Reaction With Steam</td><td colspan="5">No Reaction with Water or Steam</td></tr><tr><td>Reaction With Acids</td><td colspan="4">Violent Reaction With dilute acids</td><td colspan="4">React With dilute acids</td><td colspan="2">Slow Reaction</td><td colspan="5">No Reaction With dilute acids</td></tr></table>	Metal	Potassium	Sodium	Lithium	Calcium	Magnesium	Aluminium	Zinc	Iron	Tin	Lead	Copper	Mercury	Silver	Gold	Platinum	Reaction With Oxygen	Burn In Oxygen to Form Metal Oxide								Slowly React With Oxygen		No Reaction With Oxygen					Reaction With Water	Fast Reaction With Water				Slow Reaction With Water				Faster Reaction With Steam		No Reaction with Water or Steam					Reaction With Acids	Violent Reaction With dilute acids				React With dilute acids				Slow Reaction		No Reaction With dilute acids				
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16	D	73	<table><tr><th>Method</th><th colspan="4">Electrolysis</th><th colspan="4">Heat With Carbon</th><th colspan="4">Heat Alone</th></tr><tr><td rowspan="3">Metals Made This Way</td><td>Potassium</td><td colspan="3">Sodium</td><td>Zinc</td><td colspan="3">Iron</td><td colspan="4" rowspan="3"><table><tr><td>Mercury</td><td>Silver</td></tr><tr><td>Gold</td><td>Platinum</td></tr></table></td></tr><tr><td>Lithium</td><td colspan="3">Calcium</td><td>Nickel</td><td colspan="3">Tin</td></tr><tr><td>Magnesium</td><td colspan="3">Aluminium</td><td>Lead</td><td colspan="3">Copper</td></tr><tr><td>Reason</td><td colspan="4">most reactive metals</td><td colspan="4">medium reactive metals</td><td colspan="4">least reactive metals</td></tr></table>	Method	Electrolysis				Heat With Carbon				Heat Alone				Metals Made This Way	Potassium	Sodium			Zinc	Iron			<table><tr><td>Mercury</td><td>Silver</td></tr><tr><td>Gold</td><td>Platinum</td></tr></table>				Mercury	Silver	Gold	Platinum	Lithium	Calcium			Nickel	Tin			Magnesium	Aluminium			Lead	Copper			Reason	most reactive metals				medium reactive metals				least reactive metals								
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17	B	55	① $\text{H}_2\text{O(l)} + \text{SO}_3^{2-}(\text{aq}) \rightarrow \text{SO}_4^{2-}(\text{aq}) + 2\text{H}^+(\text{aq}) + 2\text{e}^-$ ② $\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Fe}^{2+}(\text{aq})$ ① $\text{H}_2\text{O(l)} + \text{SO}_3^{2-}(\text{aq}) \rightarrow \text{SO}_4^{2-}(\text{aq}) + 2\text{H}^+(\text{aq}) + 2\text{e}^-$ ② × 2 $2\text{Fe}^{3+}(\text{aq}) + 2\text{e}^- \rightarrow 2\text{Fe}^{2+}(\text{aq})$ Add ① + ②' (cancelling e⁻) $\text{H}_2\text{O(l)} + \text{SO}_3^{2-}(\text{aq}) + 2\text{Fe}^{3+}(\text{aq}) \rightarrow \text{SO}_4^{2-}(\text{aq}) + 2\text{H}^+(\text{aq}) + 2\text{Fe}^{2+}(\text{aq})$																																																																

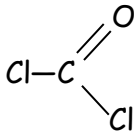
18	B	75	Magnesium Aluminium Zinc Iron Nickel Tin Lead Copper Silver A B C D
19	B	43	$ \begin{array}{c} \text{O} \quad \text{O} \quad \quad \quad \text{O} \quad \text{O} \\ \parallel \quad \parallel \quad \quad \quad \parallel \quad \parallel \\ -\text{C}-\text{C}_6\text{H}_4-\text{C}-\text{O}-(\text{CH}_2)_2-\text{O}-\text{C}-\text{C}_6\text{H}_4-\text{C}- \\ \end{array} $ $ \begin{array}{c} \text{H}_2\text{O} \quad \downarrow \text{hydrolysis} \\ \text{O} \quad \text{O} \quad \quad \quad \text{O} \quad \text{O} \\ \parallel \quad \parallel \quad \quad \quad \parallel \quad \parallel \\ \text{HO}-\text{C}-\text{C}_6\text{H}_4-\text{C}-\text{OH} + \text{H}-\text{O}-(\text{CH}_2)_2-\text{O}-\text{H} + \text{HO}-\text{C}-\text{C}_6\text{H}_4-\text{C}-\text{OH} \end{array} $
20	D	33	☒ A addition: molecule added across a C=C double bond ☒ B displacement: higher up metals displace lower down ions from compounds ☒ C neutralisation: Hydrogen H ⁺ ions react to become water H ₂ O ☒ D precipitation: insoluble substance formed when two solutions are mixed
			$\begin{array}{c} \text{NC}_{12} \quad \text{H} \\ \quad \\ \text{C} = \text{C} \\ \quad \\ \text{C} \quad \text{C} \end{array}$ $\begin{array}{c} \text{CH}_3 \quad \text{H} \\ \quad \\ \text{C} = \text{C} \\ \quad \\ \text{C} \quad \text{C} \end{array}$ $\begin{array}{c} \text{CH}_3 \quad \text{H} \\ \quad \\ \text{C} = \text{C} \\ \quad \\ \text{C} \quad \text{C} \end{array}$ + $\begin{array}{c} \text{NC}_{12}\text{H}_8 \\ \\ \text{H} \quad \text{H} \\ \quad \\ \text{C} \quad \text{C} \\ \quad \\ \text{H} \quad \text{H} \end{array}$ $\begin{array}{c} \text{NC}_{12}\text{H}_8 \\ \\ \text{H} \quad \text{H} \\ \quad \\ \text{C} \quad \text{C} \\ \quad \\ \text{H} \quad \text{H} \end{array}$ $\begin{array}{c} \text{Hydr} \quad \text{H} \\ \quad \\ \text{C} \quad \text{C} \\ \quad \\ \text{H} \quad \text{H} \end{array}$

2014 National 5 Chemistry Marking Scheme

Long Qu	Answer	Reasoning																								
1a	Repulsion/deflection by (positive) nucleus	Positive (alpha) particles mainly travel straight through the layer of gold. Some of the positive (alpha) particles travel close to the nuclei of the gold atoms. The nuclei of the gold atoms are also positive and deflect the passing positive (alpha) particles by repulsion of positives charges.																								
1b(i)	<table><tr><td>79</td></tr><tr><td>79</td></tr><tr><td>118</td></tr></table>	79	79	118	Number of protons = atomic number = 79 Number of electrons = atomic number - charge = 79 - 0 = 79 Number of neutrons = mass no - atomic no. = 197 - 79 = 118																					
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1b(ii)	Answer from:	Same atomic number but different mass number number of protons number of neutrons																								
2a	<table><tr><td>Covalent Network</td></tr><tr><td>Ionic Lattice</td></tr><tr><td>Metallic Lattice</td></tr><tr><td>Discrete Covalent Molecular</td></tr></table>	Covalent Network	Ionic Lattice	Metallic Lattice	Discrete Covalent Molecular	<table><tr><th>Bonding Type</th><th>Features of Bonding Type</th></tr><tr><td>Covalent Network</td><td>Substances with covalent bonding do not conduct electricity in any state. Covalent networks have very high melting points as covalent bonds in network must be broken before melting can take place</td></tr><tr><td>Ionic Lattice</td><td>Ionic compounds are all solids at room temperature with high mpt. Ionic compounds do not conduct as solids as ions are not free to move but conduct electricity when molten or in solution as the ions become free to move.</td></tr><tr><td>Metallic Lattice</td><td>All metallic substances conduct as solids or liquids. Metals have a range of melting point ranging from low to high e.g. mercury mpt = -39°C and iron mpt=1538°C</td></tr><tr><td>Discrete Covalent Molecular</td><td>Substances with covalent bonding do not conduct electricity in any state. As there are weak bonds between molecules, melting and boiling points are low as there are no strong bonds to be broken between molecules.</td></tr></table>	Bonding Type	Features of Bonding Type	Covalent Network	Substances with covalent bonding do not conduct electricity in any state. Covalent networks have very high melting points as covalent bonds in network must be broken before melting can take place	Ionic Lattice	Ionic compounds are all solids at room temperature with high mpt. Ionic compounds do not conduct as solids as ions are not free to move but conduct electricity when molten or in solution as the ions become free to move.	Metallic Lattice	All metallic substances conduct as solids or liquids. Metals have a range of melting point ranging from low to high e.g. mercury mpt = -39°C and iron mpt=1538°C	Discrete Covalent Molecular	Substances with covalent bonding do not conduct electricity in any state. As there are weak bonds between molecules, melting and boiling points are low as there are no strong bonds to be broken between molecules.										
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2b	Electrons are delocalised	Graphene is made of carbon. Carbon has 4 outer electrons but in graphene only 3 of these electrons are used up in covalent bonds. The 4 th electron is delocalised and able to jump from carbon atom to carbon atom allowing the conduction of electricity across grapheme.																								
3a	Humans cannot store potassium	Problem Solving: Retrieval of information from written passage																								
3b	0.022 mol	From passage: 0.86g of potassium in 100g of raisins $\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{0.86}{39} = 0.022 \text{ mol}$																								
3c	Lilac	<table><tr><td>Element</td><td>Barium</td><td>Calcium</td><td>Copper</td><td>Lithium</td><td>Potassium</td><td>Sodium</td><td>Strontium</td></tr><tr><td>Ion</td><td>Ba²⁺</td><td>Ca²⁺</td><td>Cu²⁺</td><td>Li⁺</td><td>K⁺</td><td>Na⁺</td><td>Sr²⁺</td></tr><tr><td>Flame Colour</td><td>green</td><td>orange-red</td><td>blue-green</td><td>red</td><td>lilac</td><td>yellow</td><td>red</td></tr></table>	Element	Barium	Calcium	Copper	Lithium	Potassium	Sodium	Strontium	Ion	Ba ²⁺	Ca ²⁺	Cu ²⁺	Li ⁺	K ⁺	Na ⁺	Sr ²⁺	Flame Colour	green	orange-red	blue-green	red	lilac	yellow	red
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3d	KNO ₃	<table><tr><td>Write down Valency below each ion's symbol</td><td>Put in Cross-over Arrows</td><td>Follow arrows and cancel down to get formula</td></tr><tr><td> KNO₃⁻ 11 </td><td> KNO₃⁻ </td><td>KNO₃</td></tr></table>	Write down Valency below each ion's symbol	Put in Cross-over Arrows	Follow arrows and cancel down to get formula	KNO₃⁻ 11	KNO₃⁻ 	KNO ₃																		
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4b	Addition	<table><tr><th>Polymerisation</th><th>Description</th></tr><tr><td>Addition</td><td>C=C double bonds in monomers join together to form a long chain of C-C single bonds.</td></tr><tr><td>Condensation</td><td>Water molecules are removed as monomers join together to make larger polymer.</td></tr></table>	Polymerisation	Description	Addition	C=C double bonds in monomers join together to form a long chain of C-C single bonds.	Condensation	Water molecules are removed as monomers join together to make larger polymer.																		
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5b	$\frac{1}{4}$	<table><tr><td>Time (days)</td><td colspan="2">Fraction</td></tr><tr><td>0</td><td>1</td><td>100%</td></tr><tr><td>8</td><td>$\frac{1}{2}$</td><td>50%</td></tr><tr><td>16</td><td>$\frac{1}{4}$</td><td>25%</td></tr></table>	Time (days)	Fraction		0	1	100%	8	$\frac{1}{2}$	50%	16	$\frac{1}{4}$	25%								
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5c	Sodium	Mass number of X = $28 - 4 = 24$ Atomic number of X = $13 - 2 = 11$ ∴ Element 11 = sodium																				
6	Open Question Answer to include:	<table><tr><td>3 mark answer</td><td>2 mark answer</td><td>1 mark answer</td></tr><tr><td>Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr></table>	3 mark answer	2 mark answer	1 mark answer	Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.														
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7a(i)	Haber Process	Nitrogen + Hydrogen $\xrightarrow{\text{iron catalyst}}$ Ammonia																				
7a(ii)	Diagram showing:																					
7b(i)		Arrow pointing from nitrogen monoxide produced by the absorber at the bottom of diagram leading to the nitrogen monoxide produced by the reactor.																				
7b(ii)	Water or H ₂ O	Nitric acid is made by dissolving nitrogen dioxide in water to form nitric acid. Some nitrogen monoxide is also formed during the reaction which is recycled back into the process.																				
7c(i)	Neutralisation	Neutralisation: Acid + Metal Hydroxide $\longrightarrow$ Salt + Water																				
7c(ii)	Answer to include:	Place potassium nitrate solution in an evaporation basin and boil dry using a Bunsen burner																				
8a	One answer from:	Perfumes, flavourings or solvents																				
8b(i)	hydroxyl	The -OH group of the hydroxyl group.																				
8b(ii)	Any structure from:	Correct C₅H₁₁OH diagram of: <table><tr><td>pentan-1-ol</td><td>pentan-2-ol</td><td>pentan-3-ol</td></tr><tr><td colspan="2">2-methylbutan-1-ol</td><td>2,2-dimethylpropan-1-ol</td></tr><tr><td colspan="2">2-methylbutan-2-ol</td><td>3-methylbutan-2-ol</td></tr></table> NB: diagram must be different from 3-methylbutan-1-ol in question and not a redrawing of same 3-methylbutan-1-ol structure.	pentan-1-ol	pentan-2-ol	pentan-3-ol	2-methylbutan-1-ol		2,2-dimethylpropan-1-ol	2-methylbutan-2-ol		3-methylbutan-2-ol											
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2-methylbutan-1-ol		2,2-dimethylpropan-1-ol																				
2-methylbutan-2-ol		3-methylbutan-2-ol																				

8b(iii)	$C_nH_{2n+1}COOH$ or $C_nH_{2n+1}CO_2H$	$\begin{array}{c} \text{H}-\text{C}=\text{O} \\ \\ \text{OH} \end{array}$ methanoic acid $HCOOH$ where $n=0$: $C_nH_{2n+1}COOH$	$\begin{array}{c} \text{H} & \text{H} \\ & \\ \text{H}-\text{C}-\text{C}=\text{O} \\ & \\ \text{H} & \text{OH} \end{array}$ ethanoic acid CH_3COOH where $n=1$: $C_nH_{2n+1}COOH$	$\begin{array}{c} \text{H} & \text{H} & \text{H} \\ & & \\ \text{H}-\text{C}-\text{C}-\text{C}=\text{O} \\ & & \\ \text{H} & \text{H} & \text{OH} \end{array}$ propanoic acid C_2H_5COOH where $n=2$: $C_nH_{2n+1}COOH$																												
8c	X= ethanol Y= propanoic acid	Problem Solving: Deduction from information in a table																														
9a	<table><tr><td>✓</td></tr><tr><td></td></tr><tr><td>✓</td></tr><tr><td></td></tr><tr><td></td></tr></table>	✓		✓			<table><tr><td>They have similar chemical properties</td><td>✓</td><td>Propane and butane are both members of the alkane family and have the same chemical properties.</td></tr><tr><td>They have the same molecular formula</td><td></td><td>Propane is C_3H_8 and Butane is C_4H_{10}</td></tr><tr><td>The have the same general formula</td><td>✓</td><td>All alkanes have the general formula C_nH_{2n+2}</td></tr><tr><td>They have the same physical properties</td><td></td><td>Alkanes have gradually changing physical properties e.g. bpt of propane = $-42^{\circ}C$ and bpt of butane = $-1^{\circ}C$</td></tr><tr><td>The have the same formula mass</td><td></td><td>gfm propane C_3H_8 = 44g gfm butane C_4H_{10} = 58g</td></tr></table>	They have similar chemical properties	✓	Propane and butane are both members of the alkane family and have the same chemical properties.	They have the same molecular formula		Propane is C_3H_8 and Butane is C_4H_{10}	The have the same general formula	✓	All alkanes have the general formula C_nH_{2n+2}	They have the same physical properties		Alkanes have gradually changing physical properties e.g. bpt of propane = $-42^{\circ}C$ and bpt of butane = $-1^{\circ}C$	The have the same formula mass		gfm propane C_3H_8 = 44g gfm butane C_4H_{10} = 58g										
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9b	Answer to include:	Butane C_4H_{10} has more carbons than propane C_3H_8 . The larger the molecules the higher the boiling point due to (Higher would be London Dispersal Forces but this is NAT5)																														
9c	3135	<table><tr><td>E_h</td><td>=</td><td>c</td><td>x</td><td>m</td><td>x</td><td>ΔT</td></tr><tr><td>Energy</td><td>=</td><td>specific heat capacity</td><td>x</td><td>mass</td><td>x</td><td>change in temperature</td></tr><tr><td>Energy</td><td>=</td><td>4.18</td><td>x</td><td>25</td><td>x</td><td>30</td></tr><tr><td>Energy</td><td>=</td><td>3135kJ</td><td colspan="4"></td></tr></table>			E_h	=	c	x	m	x	ΔT	Energy	=	specific heat capacity	x	mass	x	change in temperature	Energy	=	4.18	x	25	x	30	Energy	=	3135kJ				
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9d	SO_2 produced or Acid Rain gas formed	Sulphur atoms in compound will burn to form sulphur dioxide. Sulphur dioxide dissolves in rain water in the atmosphere to form sulphurous acid.																														
10a(i)	Increase in number of carbons gives increase in flash point	Problem Solving: Forming a conclusion from table of information																														
10a(ii)	48°C-50°C	<table><tr><td>Alkane</td><td>Hexane C_6H_{14}</td><td>Heptane C_7H_{16}</td><td>Octane C_8H_{18}</td><td>Nonane C_9H_{20}</td><td>Decane $C_{10}H_{22}$</td></tr><tr><td>Flash Point ($^{\circ}C$)</td><td>-23</td><td>-4</td><td>13</td><td>31</td><td>-</td></tr><tr><td>Difference</td><td></td><td>19°C</td><td>17°C</td><td>18°C</td><td>17°C → 19°C</td></tr><tr><td>Prediction</td><td>-</td><td>-</td><td>-</td><td>-</td><td>48°C-50°C</td></tr></table>				Alkane	Hexane C_6H_{14}	Heptane C_7H_{16}	Octane C_8H_{18}	Nonane C_9H_{20}	Decane $C_{10}H_{22}$	Flash Point ($^{\circ}C$)	-23	-4	13	31	-	Difference		19°C	17°C	18°C	17°C → 19°C	Prediction	-	-	-	-	48°C-50°C			
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Prediction	-	-	-	-	48°C-50°C																											
10b	99g	gfm C_9H_{20} = $(9 \times 12) + (20 \times 1) = 108 + 20 = 128g$ no of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{32}{128} = 0.25\text{mol}$ $C_9H_{20} + 14O_2 \longrightarrow 9CO_2 + 10H_2O$ 1mol 0.25mol																														

11b(i)	sodium hydroxide	Solution X is the other product of the reaction happening around the negative electrode. - Sodium ions are crossing the membrane into the right hand side - Water is splitting into H^+ ions and OH^- ions $2H^+ + 2e^- \longrightarrow H_2$ leaving OH^- ions behind																																	
11b(ii)	It is renewable or no CO_2 /greenhouse gases produced	Hydrogen H_2 gas burns cleanly to form water H_2O . Hydrogen can then be regenerated from the water making hydrogen a renewable fuel. As there is no carbon in H_2 , no poisonous carbon monoxide will be produces and no carbon dioxide, a greenhouse gas, is produced																																	
11c		<table><tr><th>Element</th><th>Group</th><th>Valency</th><th>No of Bonds Element Makes</th></tr><tr><td>Carbon</td><td>4</td><td>4</td><td>4</td></tr><tr><td>Oxygen</td><td>6</td><td>2</td><td>2</td></tr><tr><td>Chlorine</td><td>7</td><td>1</td><td>1</td></tr></table>	Element	Group	Valency	No of Bonds Element Makes	Carbon	4	4	4	Oxygen	6	2	2	Chlorine	7	1	1																	
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12a	Reduction	Metal ores are compounds of metals and the metals are in the form of positive ions. Positive ions are turned back in atoms again be reduction where electrons are gained by the positive ions e.g. $Mg^{2+} + 2e^- \rightarrow Mg$																																	
12b	70%	$\% Fe = \frac{\text{total mass of Fe}}{\text{gfm } Fe_2O_3} \times 100 = \frac{(2 \times 56)}{(2 \times 56) + (3 \times 16)} \times 100 = \frac{112}{160} \times 100 = 70\%$																																	
12c	(molten) Electrolysis	<table><tr><th>Method</th><th colspan="2">Electrolysis</th><th colspan="2">Heat With Carbon</th><th colspan="2">Heat Alone</th></tr><tr><td rowspan="4">Metals Made This Way</td><td>Potassium</td><td>Sodium</td><td>Zinc</td><td>Iron</td><td>Mercury</td><td>Silver</td></tr><tr><td>Lithium</td><td>Calcium</td><td>Nickel</td><td>Tin</td><td>Gold</td><td>Platinum</td></tr><tr><td>Magnesium</td><td>Aluminium</td><td>Lead</td><td>Copper</td><td></td><td></td></tr><tr><td>Reason</td><td colspan="2">most reactive metals</td><td colspan="2">medium reactive metals</td><td colspan="2">least reactive metals</td></tr></table>	Method	Electrolysis		Heat With Carbon		Heat Alone		Metals Made This Way	Potassium	Sodium	Zinc	Iron	Mercury	Silver	Lithium	Calcium	Nickel	Tin	Gold	Platinum	Magnesium	Aluminium	Lead	Copper			Reason	most reactive metals		medium reactive metals		least reactive metals	
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13a	$16cm^3$	Average Volume = $\frac{15.9 + 16.1}{2} = \frac{32.0}{2} = 16.0cm^3$ NB: Ignore rough titre in the calculation of the average volume as it is inaccurate																																	
13b	0.08 mol l^{-1}	no. of mol = volume $\times$ concentration = $0.016\text{litres} \times 0.1 \text{ mol l}^{-1} = 0.0016\text{mol}$ $2HCl + Na_2CO_3 \longrightarrow 2NaCl + CO_2 + H_2O$ 2mol 1mol 0.0016mol 0.0008mol concentration = $\frac{\text{no. of mol}}{\text{volume}} = \frac{0.0008 \text{ mol}}{0.01 \text{ litres}} = 0.08 \text{ mol l}^{-1}$																																	
14	Open Question Answer to include:	<table><tr><th>3 mark answer</th><th>2 mark answer</th><th>1 mark answer</th></tr><tr><td>Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr></table>	3 mark answer	2 mark answer	1 mark answer	Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.																											
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2014 National 5 Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D
1	A	91	7	1	1
2	D	3	7	6	83
3	A	51	29	8	11
4	C	2	7	89	2
5	C	9	4	75	12
6	A	59	17	10	14
7	D	7	5	6	82
8	A	88	3	2	7
9	C	2	4	87	7
10	B	1	83	9	6
11	C	14	3	77	6
12	B	20	40	11	27
13	C	2	4	88	6
14	C	20	14	47	19
15	A	75	14	6	6
16	D	7	11	8	73
17	B	28	55	10	6
18	B	12	75	10	3
19	B	27	43	7	23
20	D	30	19	18	33